

Design of Local Area DGNSS Architecture to Support UAV Networks: Optimal Integrity/Continuity Allocations and Fault Monitoring

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ABSTRACT

A Local-Area Differential GNSS (LAD-GNSS) provides high integrity and accuracy for Unmanned Aerial Vehicles (UAVs). The use of a two-way datalink to relay UAV information back to the ground station can improve the performance and integrity of the LAD-GNSS. This paper proposes modifications using two-way datalink and fully analyzes the performance of the proposed system. Key components of the proposed system in this work, include an operation requirement design, the modification of integrity monitoring, and an optimal allocation of integrity/continuity budgets.

In this paper, two operation requirements, vertical alert limit (VAL) and tolerable error limit (TEL), are derived based on the operating environment and the uncertainty of navigation. Vertical protection level (VPL) under optimal allocation and maximum ionospheric error in vertical (MIEV) computed in real time are compared to VAL and TEL respectively to determine if the system meets the integrity requirements for a given operation. This paper also describe an methodology to derive PLs for the excessive acceleration and code-carrier divergence fault scenarios which were bounded by using a maximum-allowable error in range (MERR). Using these PLs, the integrity/continuity allocations ideally are assigned dynamically to make the PLs to be identical for all fault scenarios. The results show that the VPL are reduced by about 8% when implement optimal allocation method.

1.0 INTRODUCTION

One of the most essential technologies required for the operation of Unmanned Aerial Vehicle (UAV) is reliable and accurate computation of navigation solution. A concept of local-area UAV network which utilizes Local-Area Differential GNSS (LAD-GNSS) navigation solutions has already been proposed to achieve high accuracy and reliability of navigation that are needed for autonomous UAV applications [1]. The LAD-GNSS meets the required level of integrity requirements (comparable to those of GBAS Category I – III operations) by monitoring navigation faults at both the reference station and UAV and by broadcasting integrity information to the UAV. By

utilizing the integrity information which defines navigation system error models, the UAVs can compute their conservative position error bounds (i.e., protection levels (PL)) and consequently safe separation distances in real time. The LAD-GNSS architecture for local-area UAV network proposed in previous work includes several methods of simplifying the current GBAS integrity monitoring algorithms and hardware configurations to lower cost and complexity of the system while maintaining an acceptable level of safety [1]. The performance and integrity of LAD-GNSS can also improve by designing additional modules using an airborne-to-ground datalink. The LAD-GNSS integrity monitor testbed have been installed in the KAIST campus, but those proposed modifications including ground hardware/software changes and airborne components have not been fully analyzed and implemented yet in a system design perspective.

This study develops the several key components of the LAD-GNSS architecture including operation coverages and requirements of the system, the modifications of integrity monitors in both ground and airborne subsystems, and optimal integrity/continuity allocations. In the current GBAS architecture, separate PL values for three scenarios (i.e., fault-free (H0), single reference-receiver fault (H1), and ephemeris-fault cases) are defined based on the predetermined integrity and continuity allocations. These allocations ideally could be assigned dynamically to make the PLs to be identical for all fault scenarios. The use of a two-way datalink to relay UAV information back to the ground station provides a significant advantage over the conventional GBAS and makes the real-time allocation possible. Since the LAD-GNSS ground station can be notified about the satellite geometry which the UAV utilizes for its positioning, it is possible to compute the PLs for all fault scenarios and optimally allocate the integrity and continuity budgets in order to make the PLs to be identical for all fault scenarios.

The integrity monitors of LAD-GNSS are developed using the well-established GBAS as a starting point. In this study, we implement major modifications on the ground subsystems using two-way datalink. First, by using known user geometry, the ionospheric geometry screening to mitigate worst-case ionospheric gradients is simplified. The simplified geometry screening searches only one

geometry instead of searching every possible combination of satellite geometries. In addition, the ground Dual Smoothing Ionosphere Gradient Monitor Algorithm (DSIGMA) is utilized to improve the performance of geometry screening. In this paper, Section 2 illustrates the LAD-GNSS architecture including the operation coverages, and the modifications of integrity monitoring. Section 3 explains operation requirement of LAD-GNSS and Section 4 explains the methodology of optimal integrity and continuity allocations in detail. Section 5 analyzes geometry screening to mitigate the worst-case ionospheric gradients and shows the simulation results. Section 6 concludes the paper.

2.0 LAD-GNSS ARCHITECTURE FOR UAV

The proposed LAD-GNSS architecture for UAV emphasizes low-cost, space-conserving installation and low complexity on monitoring algorithms for commercial applications while supporting high navigation integrity within the coverage airspace. Figure 1 illustrates the concept of LAD-GNSS architecture for UAV. The ground facility shown at the center is the source of LAD-GNSS corrections and integrity information as well as real-time guidance for each UAV operating within the coverage airspace. To enable safe operations of UAV in the civil airspace, coverage airspace by the system should be carefully defined. As the distance from the ground facility to an operating UAV increases, residual navigation errors which remain even after applying differential corrections increase. Thus, the error uncertainty over distance from the ground facility to UAVs should be considered when defining the coverage airspace to be safely integrated into civil airspace. The coverage airspaces of the proposed system were pre-defined based on expected UAV missions for each regions for choosing operation radius, and results of simulation for vertical position uncertainty (includes VPL and maximum ionospheric error in vertical (MIEV)) for determining operation heights. The defined coverage airspaces are listed in Table 1.

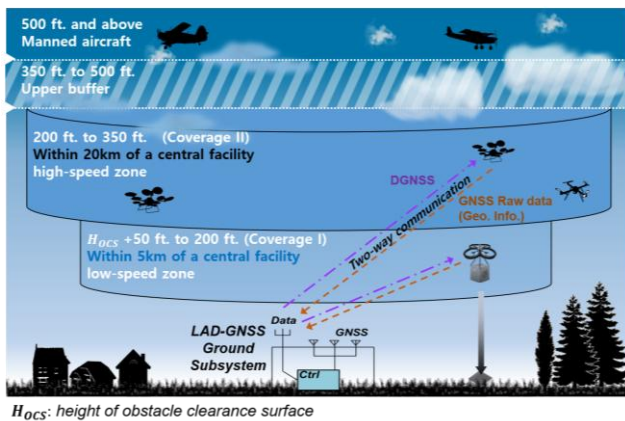


Figure 1. The concept of LAD-GNSS architecture for UAV

Table 1. Coverage airspace of LAD-GNSS

Criteria	Radius (from the ground facility)	Height
Coverage I	~ 5km	$50\text{ft} + H_{OCS} \leq h_1 < 200\text{ft}$
Coverage II	5km~20km	$200\text{ft} \leq h_2 < 350\text{ft}$

‘Coverage I’ supports low-speed UAV operations from 200 ft altitude down to an altitude of 50 ft + H_{OCS} within the radius of 5 km. The minimum altitude, 50 ft, is chosen from the simulations of VPL and MIEV at 5 km apart from the ground facility (where the worst-case VPLs and MIEVs are derived). The detail of the simulation will be discussed in Section 4 and 5. The H_{OCS} indicates the maximum height of an obstacle that can be placed along the aircraft approach path. H_{OCS} for GBAS have already been defined and are enforced at runways that have precision approaches, non-precision approaches and even visual operations [2]. H_{OCS} for UAVs should be carefully defined based on the environment where the UAV is actually flying above. One potential application proposed for the Coverage I can be UAV landing mission down to 50ft + H_{OCS} . Below the minimum height, aid of other sensors, such as a vision sensor or a laser sensor, is required for touch-down. ‘Coverage II’ is designed to support high-speed operations between 200 ft and 350 ft within the radius of 20 km from the ground station. The upper limit, 350 ft, is determined to avoid penetration of the airspace where manned aircrafts are operating (above the 500 ft) due to the vertical position uncertainty. The magnitude of the upper buffer in Figure 1 from 350 ft to 500 ft is determined in a same manner as the lower limit is defined. Only difference is that H_{OCS} is not considered for determining the upper buffer.

One of the key feature of the proposed system is to utilize a two-way datalink between the ground facility and UAVs in the navigation system which was originally used for the control and guidance function. The proposed system relays GNSS raw measurements received from UAVs to the ground facility in real-time as shown in Figure 2. The ground facility performs monitoring, correction generation and PL computation. After all the process at the ground facility, UAVs only receive differential corrections and compute their corrected position solution on onboard. PLs and alert messages are transmitted to the ground controller for safety assured guidance. The intent of this new approach is to rely on major computations at the ground facility rather than at the onboard receivers to reduce computation burden on onboard for an easiness of implementation. This modified configuration provides significant benefits to fault monitoring algorithms and PL computation algorithms by having known user satellite geometries.

Figure 2 describes the modified configuration and components of LAD-GNSS. Three ground reference receivers are connected to low-cost antennas. Siting constraints for most commercial applications will not have room to spread ground antennas over hundreds of meters.

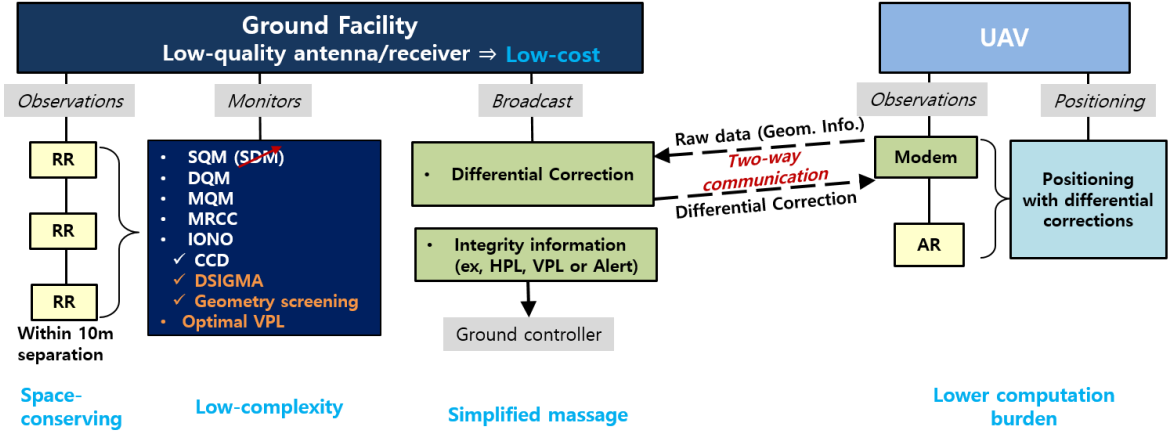


Figure 2. The configuration and components of LAD-GNSS architecture

In this system, antennas are separated from each other by 10 meters or less. Low-quality antennas and relatively short antenna separation increase the impact of multipath errors. To mitigate performance degradation due to the hardware restriction, and to take advantages from the new system configuration, this study develops the several key components of the LAD-GNSS architecture including optimal integrity allocations and the modifications of integrity monitors. This would reduce PLs compared to the original PLs computed from the pre-allocated integrity budgets. The algorithms for optimal PL computation is introduced in Section 4. In addition to the optimal allocation of integrity budget, modification of monitors can potentially improve the overall performance. Geometry screening, which is implemented to mitigate the integrity risk due to anomalous ionosphere gradients with a code-carrier divergence (CCD) monitor and DSIGMA, can be simplified based on known user geometry. The simplified geometry screening monitor operates to compute MIEV in real-time. Since there is no need to consider all the possible user geometries, both reduced computation load and improved availability would be the benefits from the modified geometry screening. The concept of modified geometry screening and MIEV simulations will be discussed in Section 5. Lastly, Signal Deformation Monitor (SDM) also can be eliminated by selecting ground and airborne GNSS receivers that are either the same or very similar. This would cancel out the resulting error when users apply their differential corrections [1].

3.0 OPERATION REQUIREMENT FOR LAD-GNSS FOR UAV

3.1 Coverage I and II Vertical Alert Limits

For the GBAS, the PLs are compared to alert limit in each position axis of interest to determine if the system meets the integrity and continuity requirements for a given operation. A VAL of 10 m for GBAS CAT I has been established based on the vertical error characteristics of Instrument Landing System (ILS) [3]. Since the Coverage II of LAD-GNSS has the same low limit as GBAS CAT I

decision height of 200 ft, the UAV operating in Coverage II uses VAL of 10 m to determine if it meets the system requirements. However, the UAV operating in Coverage I may need a different value from CAT-I VAL because it is flying as low as 50 ft.

In this paper, the VAL of Coverage I is analyzed in a manner similar to GBAS CAT III. The analysis focuses on Total System Error (TSE) as composed of the sum of Navigation Sensor Error (NSE) and Flight Technical Error (FTE). The key requirement for vertical TSE performance of GBAS CAT III are assuring touchdown no shorter than 200 ft from threshold, and no longer than 2700 ft [3]. Similar to the GBAS CAT III, the performance criteria of Coverage I of LAD-GNSS are stated in terms of height of UAV low than H_{OCS} . Corresponding to these requirements, the analysis is divided into the fault-free NSE case, followed by the faulted NSE case. The allowed risk for fault-free case is 10^{-6} and the allowed risk for faulted case is 10^{-5} [3, 4]. The paper treats only the fault-free NSE case because it has a tighter requirement than that of faulted case.

To facilitate the analysis, the simplifying assumptions used in this analytical derivation. It will assumed that the relationship between vertical NSE and VAL at the worst geometry is expressed as

$$\sigma_{NSE,v}(m) = \frac{VAL(m)}{k_{FFMD}}; \quad k_{FFMD} = 5.81 \quad (1)$$

where k_{FFMD} is a multiplier defined to ensure a specified probability of fault-free missed detection. The vertical TSE is normally distributed with zero mean and standard deviation ($\sigma_{TSE,v}$) given by

$$\sigma_{TSE,v} = \sqrt{\sigma_{NSE,v}^2 + \sigma_{FTE,v}^2} \quad (2)$$

It will also be assumed that if TSE is zero, the height of UAV will be 50 ft from H_{OCS} . Therefore, the probability that the height of UAV will be low than H_{OCS} is given by

$$P_{risk} = \int_{-\infty}^{-50} \frac{1}{\sqrt{2\pi}\sigma_{TSE,v}} e^{-\frac{x^2}{2\sigma_{TSE,v}^2}} dx \quad (3)$$

Figure 3 shows a plot of the fault-free risk of Coverage I versus VAL for values of $\sigma_{FTE} = 3\text{ m}$, $\sigma_{FTE} = 2.7\text{ m}$, $\sigma_{FTE} = 2\text{ m}$, and $\sigma_{FTE} = 1.5\text{ m}$. Note that for $\sigma_{FTE} = 2.7\text{ m}$, the risk for VAL of 10 m exactly meets the risk requirement of 10^{-6} . The previous works have shown that the σ_{FTE} of the multi-copter UAV is sub-meter level [5]. Based on the conservative value of $\sigma_{FTE} (= 1.5\text{ m})$, the VAL of 10 m for Coverage I is determined.

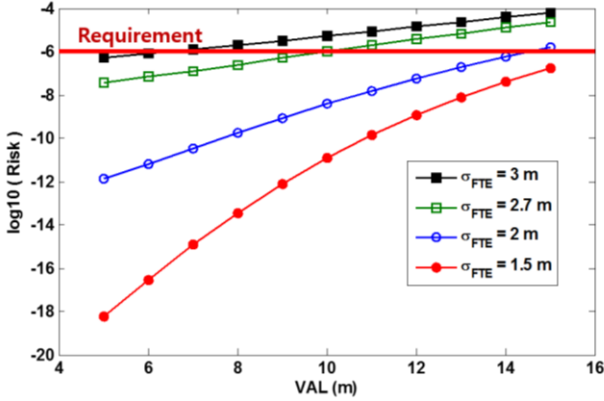


Figure 3. Fault-Free Risk for Coverage I

3.2 Coverage I and II Tolerable Error Limit

One of the most challenging hazards to mitigate is position error due to rare ionospheric anomalies. Such anomalies can produce large spatial gradients of the ionospheric delay. Under general flight conditions, it is very difficult to guarantee that the worst-case undetected ionospheric spatial gradient is bounded by the protection levels computed by aircraft. To mitigate this threat, GBAS CAT I has been implement ionospheric geometry screening method. The geometry screening is to compute the worst-case undetected position error (denoted as MIEV) in the ground system and increase one or more broadcast parameters that affect the airborne protection levels so that satellite geometries potentially giving hazardous errors are made unavailable. In this methodology, the unusable satellite geometries is determined by comparing the MIEV of geometries with tolerable error limit (TEL) in the vertical direction derived from the obstacle clearance surface for precision approach [6]. The TEL for manned aircrafts is derived as a combination of the ideal height of the aircraft in approaching phase, 95% percentile of the fault-free TSE and the obstacle clearance level near the airport according to (4) [7].

$$TEL(D) = H_{AC}(D) - TSE_{95\%}(D) - H_{OCS}(D) \quad (4)$$

where H_{AC} is the height of the aircraft on ideal glide path at a distance D from the runway threshold and H_{OCS} is the height of obstacle clearance surface. $TSE_{95\%}$ is given by

$$TSE_{95\%} = 1.96 \sqrt{\sigma_{NSE,v}^2 + \sigma_{FTE,v}^2}; \quad \sigma_{NSE,v} = \frac{VAL}{k_{FFMD}} \quad (5)$$

For the LAD-GNSS for UAV, the MIEV computed by using UAV satellite geometry is compared to TEL in real time. The TEL for Coverage I and II of LAD-GNSS is established in similar manner to Eq. (4), with the exception that H_{AC} is substituted with low limit of Coverage I and II. The TELs for Coverage I and II are shown in Eq. (6).

$$TEL_{LAD-GNSS} = \begin{cases} 50 \text{ ft.} - TSE_{95\%}(d), & \text{if } d < 5 \text{ km} \\ 200 \text{ ft.} - TSE_{95\%}(d) - H_{OCS}(d), & \text{if } 5 \text{ km} < d < 20 \text{ km} \end{cases} \quad (6)$$

where d is distance between the ground subsystem and the UAV.

4.0 OPTIMAL ALLOCATION

In this section, we describe methodology for optimally allocating the integrity/continuity budgets. As mentioned Section 1, using the known satellite geometry of UAV, the ground subsystem can compute the PLs for all fault scenarios and optimally allocate the integrity and continuity budgets in order to make the PLs to be identical for all fault scenarios. As the allocations are optimized with the PLs for each fault mode, the PLs for fault modes that were bounded by using a Maximum-allowable Error in Range (MERR) method should be newly derived.

Figure 4 show the integrity fault tree of the system applied the optimal allocation. The total system signal in space integrity risk requirement of 2×10^{-7} per 150 sec is allocated. The total system integrity requirement remains unchanged from that of the GBAS CAT I. Instead, sub-allocations are adjusted by the inclusion of the optimal allocation. In the re-designed integrity fault tree, the integrity risk of single deformation, low signal level, and all other failures is pre-allocated. This is because that the signal deformation error is canceled out when ground subsystem and UAV use the receivers that are either the same or very similar, and thus the small value (1×10^{-8} per 150 sec) is allocated to the signal deformation integrity risk. In case of the low signal level integrity risk, the 1×10^{-8} per 150 sec is also pre-allocated because the bound of low signal level is determined based on the impact on the performance of other monitors, not on MERR. Finally, as the all other failures cannot be bounded by MERR or by nominal VPL, the integrity risk same as GBAS CAT I is allocated.

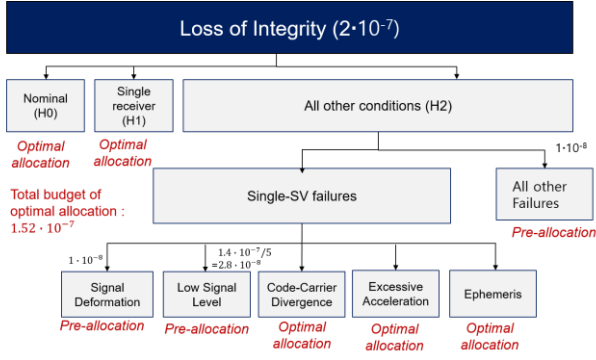


Figure 4. Integrity risk allocation of LAD-GNSS for UAV

4.1 Derivation of Excessive Acceleration Protection Levels

The excessive acceleration (EA) threat is a fault condition in a GNSS satellite that causes the range measurements to accelerate excessively. The most probable causes would be GNSS clock anomalies or ground receiver failures by verifying the consistency of the pseudorange and carrier-phase measurements over the last few epochs. The monitor is designed to detect EA fault by comparing the estimation of range acceleration error with threshold [8, 9]. The range acceleration error at epoch k is estimated by

$$acc(k) = \frac{\rho_{c,i}(k+2) - \rho_{c,i}(k) + \rho_{c,i}(k-2)}{(2T)^2} \quad (7)$$

where $\rho_{c,i}$ is smoothed pseudorange correction for satellite i , and T is measurement update time interval, for LAD-GNSS ground subsystem this value is 0.5 seconds. The effective differential ranging error ($\delta\rho_{EA}$) due to an acceleration (acc) for satellite i is:

$$\delta\rho_{EA,i} = 0.5 \cdot \tau \cdot (\tau + T) \cdot acc \quad (8)$$

In this equation, the nominal time delay $\tau = 1.5(ground) + 1(air) = 2.5$ seconds [9].

The VPL under EA hypothesis for satellite i consists of two components, as shown in Eq. (9). First, the position error bound (μ_{EA}) is computed based on the bias induced by the satellite fault. Second, the random error term is computed using a sigma-multiplier (k_{HMI}), which considers that monitors miss-detect such faults.

$$VPL_{EA}(i) = \mu_{EA,i} + k_{HMI,EA} \sum_{i=1}^n S_{vert,i}^2 \sigma_i^2 \quad (9)$$

where k_{HMI} is a multiplier defined to ensure a specified probability of missed detection, and σ_i is standard deviation of differential ranging error for satellite i , and S_{vert} is the row of the weighted-least-squares projection matrix corresponding to the vertical position state [3]. In this paper, the position error bound (μ_{EA}) for EA is derived by using minimum detectable error (MDE) concept [3] and Eq (8).

$$\mu_{EA,i} = S_{vert,i} \cdot 0.5 \cdot \tau \cdot (\tau + T) \cdot (k_{cont,EA} + k_{HMI,EA}) \sigma_{t_{EA}} \quad (10)$$

Here, k_{cont} is a multiplier defined to ensure a specified probability of fault-free alarm, and $\sigma_{t_{EA}}$ is the standard deviation of the nominal test statistic variation (acc). Considering the model of the test statistics as in equation Eq. (7), the standard deviation of statistics is

$$\sigma_{t_{EA}} = \sigma_{noise,PR} \sqrt{1^2 + 2^2 + 1^2} \quad (11)$$

where $\sigma_{noise,PR}$ is standard deviation of smoothed pseudorange noise. The standard deviation of statistics with 3 reference receivers (RR) and the lowest carrier-to-noise ratio at 32 dB-Hz is $1.76/\sqrt{3}$ cm.

4.2 Derivation of Code-Carrier Divergence Protection Levels

The code-carrier divergence monitor is used to detect ionospheric storms and ensure that the divergence of code and carrier for any given satellite is sufficiently small. The divergence rate is the estimated change rate of ionosphere delay, I , over time. The input to the divergence rate estimator, z , is the raw code minus carrier measurement. The discrete time equations for the divergence rate estimator are given below [10],

$$\begin{aligned} d_1(k) &= \frac{\tau_{d1} - T}{\tau_{d1}} d_1(k-1) + \frac{1}{\tau_{d1}} (z(k) - z(k-1)) \\ Div(k) &= \frac{\tau_{d2} - T}{\tau_{d2}} Div(k-1) + \frac{T}{\tau_{d2}} d_1(k) \end{aligned} \quad (12)$$

where τ_d is the time constant whose values are nominally set at $\tau_{d1} = \tau_{d2} = 30$ sec, and T is the output interval of the raw measurements. Given a divergence failure with a CCD rate Div and time onset $t = 0$, a differential ranging error ($\delta\rho_{CCD,i}$) due to an divergence rate for satellite i is [10],

$$\delta\rho_{CCD,i} = |e_g(t - \max[t_{g,0}, 0]) - e_a(t - \max[t_{a,0}, 0])| \quad (13)$$

where e_g is ranging error due to Div at ground relative to ground steady state, and e_a is ranging error due to Div at aircraft relative to ground steady state, $t_{g,0}$ and $t_{a,0}$ is ground filter start time and airborne filter start time respectively. Figure 5 shows the worst-case differential ranging error at Div of 0.004 m/s. Note that prior research [10] suggests that a nominal value of σ_{Div} is 0.004 m/s. The worst-case occurs when the ground filter has already converged and aircraft filter has just started. For satellite newly acquired, the pseudorange is used after two smoothing time constant intervals, or 200 sec, to reach steady-state. Therefore, the worst differential range error at σ_{Div} of 0.004 m/s is 0.074 as shown in Figure 5. Such like VPL for EA, the VPL for CCD is derived by using worst differential range error at σ_{Div} of 0.004 m/s and MDE

concept.

$$VPL_{CCD}(i) = S_{vert,i} \cdot (k_{cont,CCD} + k_{HMI,CCD}) \cdot 0.074 + k_{HMI,CCD} \sum_{i=1}^n S_{vert,i}^2 \sigma_i^2 \quad (14)$$

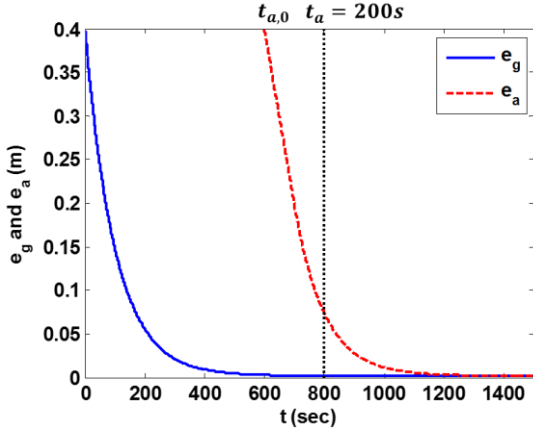


Figure 5. Ranging error due to Div of 0.004 m/s relative to ground subsystem steady state

4.3 Methodology of Optimal Allocation

The optimal problem for finding the minima of PLs subject to the constraints that the sum of allocations of each fault mode equals the total risks is solved by using a Lagrangian function [11]. In this methodology, all of the VPLs are expressed as function of the continuity and integrity risk as shown in Eq. (15)

$$VPL_i \left(P_{cont,i}, \frac{PHMI_i}{P_{ap,i}} \right) = M_i(P_{cont,i}) + L_i \left(\frac{PHMI_i}{P_{ap,i}} \right) \quad (15)$$

where $P_{cont,i}$ and $PHMI_i$ are continuity and integrity allocated to hypothesis i , $P_{ap,i}$ is prior probability of hypothesis i . Once the VPLs are organized into the above expression, two-dimensional optimization problem can be written as

$$\begin{aligned} & \text{Minimize} \quad \max_i \left[M_i(P_{cont,i}) + L_i \left(\frac{PHMI_i}{P_{ap,i}} \right) \right] \\ & \text{subject to} \quad \sum_{i=0}^{N_{mod}-1} P_{cont,i} = P_{cont} \\ & \text{and} \quad \sum_{i=0}^{N_{mod}-1} PHMI_i = PHMI \end{aligned} \quad (16)$$

where P_{cont} and $PHMI$ are total continuity and total integrity risk respectively, and N_{mod} is the number of hypotheses. Using the result of Section 4.1 and 4.2, the VPLs for LAD-GNSS can be expressed as

$$\begin{aligned} VPL_{H0} &= k_{HMI,H0} \sum_{i=1}^n S_{vert,i}^2 \sigma_i^2 \\ VPL_{H1}(j) &= k_{HMI,H1} \sum_{i=1}^n S_{vert,i}^2 \sigma_{H1,i}^2 + |B_{vert,j}| \\ VPL_e(i) &= k_{cont,H0} \cdot \left(\frac{|S_{vert,i}| |x|}{R_i} \sigma_{t_e} \right) \\ &+ k_{HMI,H0} \left(\frac{|S_{vert,i}| |x|}{R_i} \sigma_{t_e} + \sum_{i=1}^n S_{vert,i}^2 \sigma_i^2 \right) \\ VPL_{EA}(i) &= k_{cont,EA,i} \cdot (|S_{vert,i}| \cdot 0.5 \cdot \tau \cdot (\tau + T) \sigma_{t_{EA}}) \\ &+ k_{HMI,H0} \left(|S_{vert,i}| \cdot 0.5 \cdot \tau \cdot (\tau + T) \sigma_{t_{EA}} + \sum_{i=1}^n S_{vert,i}^2 \sigma_i^2 \right) \\ VPL_{CCD}(i) &= k_{cont,H0} \cdot (|S_{vert,i}| \cdot 0.074) \\ &+ k_{HMI,H0} \cdot (|S_{vert,i}| \cdot 0.074 + \sum_{i=1}^n S_{vert,i}^2 \sigma_i^2) \end{aligned} \quad (17)$$

where σ_{H1} is standard deviation under single reference-receiver fault, and B_{vert} is B-value converted to vertical position error, and x is the displacement vector between the reference station and aircraft antennas, and R_i is the distance (scalar) to satellite i . It is practical at this point to change notations and variables so that the notations are simpler. In Eq. (17), we can change the terms multiplied by continuity, the terms multiplied by integrity, and B-value into θ , ψ , and β , respectively. The problem to be solved is then

$$\begin{aligned} & \text{Minimize} \quad \max_i [k_{HMI,i} \sigma_i + k_{cont,i} \theta_i + \beta_i] \\ & \text{subject to} \quad \sum_{i=0}^{N_{mod}-1} 2a_i Q(-k_{cont,i}) = P_{cont} \\ & \text{and} \quad \sum_{i=0}^{N_{mod}-1} 2P_{ap,i} Q(-k_{HMI,i}) = PHMI \end{aligned} \quad (18)$$

where Q is unit Gaussian cumulative distribution function, and a_0 is zero and a_i is one. According to [11], by allocating the continuity budget arbitrarily, the optimal problem can be simplified to one-dimensional search space. In this suboptimal scheme, the parameter $k_{cont,i}$ is therefore no longer a variable, and we can define

$$\beta'_i = k_{cont,i} \theta_i + \beta_i \quad (19)$$

The problem to be solved is then

$$\begin{aligned} & \text{Minimize} \quad \max_i [k_{HMI,i} \sigma_i + \beta'_i] \\ & \text{subject to} \quad \sum_{i=0}^{N_{mod}-1} 2P_{ap,i} Q(-k_{HMI,i}) = PHMI \end{aligned} \quad (20)$$

In this paper, we implement the suboptimal scheme because the optimal allocation of the continuity offers only a minor improvement while increasing the complexity of algorithm and computation load [11].

From the suboptimal scheme described above and the VPLs derived in Section 4.1 and 4.2, simulations of a VPL under optimal allocation were conducted. For the simulations, the continuity risks were allocated the same as GBAS CAT-I and the standard RTCA 24 constellation was used to generate visible satellite geometries. The satellite geometries were simulated every 5 minutes (which gives 288 epochs for one day). We also assumed that $\sigma_{t_{EA}}$ is 0.99 cm/s^2 at C/N_0 of 32 db-HZ and σ_{Div} is 0.004 m/s . For the standard deviation of ground error (σ_{pr_gnd}), Ground Accuracy Designator (GAD)-B model with $M=1$ was used. The reason for using $M=1$ is that the effect of mitigating the multipath error is not expected in an environment where the separation distances between the antennas are narrow. The standard deviation of airborne error (σ_{air}) was computed by using the Airborne Accuracy Designator (AAD)-B model for simulation.

Figure 6 shows the result VPLs for Coverage I and II when the UAV is operating 5 km apart from the ground subsystem with a constant speed of 15 m/s in Coverage I and 20 km apart from ground subsystem with a constant speed of 45 m/s in Coverage II. In this result, the VPLs for fault-free (H_0), single reference-receiver fault (H_1), ephemeris fault cases, EA-fault cases, and CCD-fault cases were computed by using the integrity risk of the GBAS CAT-I. As show in Figure 6, the VPL under optimal allocation of integrity risk, which is shown by the black curve, is slightly below the VPL H_0 . These results show that the VPLs of Coverage I and II are reduced by about 8% by using the optimal allocation. The VPL under optimal allocation of integrity is reasonably well-bounded by VAL of 10 m which is the operation requirements for Coverage I and II.

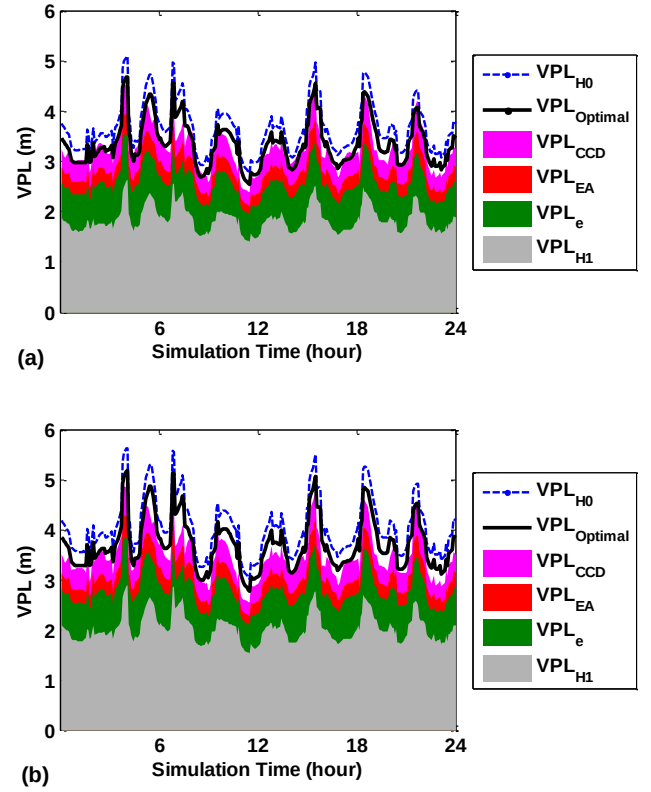


Figure 6. VPLs of LAD-GNSS for (a) UAV operating in Coverage I, and (b) UAV operating in Coverage II

5.0 MITIGATION OF IONOSPHERIC INTEGRITY RISK

In order to mitigate the integrity risk due to anomalous ionosphere gradients, several monitors including the ground CCD monitor, DSIGMA, and the simplified geometry screening monitor are implemented in LAD-GNSS. The CCD monitor and DSIGMA is capable of detecting and excluding the satellite impacted by large ionospheric gradients using divergence rate of code-carrier measurements. However, the ionosphere gradients moving with slow speed are difficult to be detected by both the ground CCD monitor and DSIGMA, which results in unacceptable large residual errors in GNSS range signals, even after applying differential corrections. Thus, the simplified geometry screening monitor operates to compute MIEV in real-time.

In order to assess impacts of anomalous ionosphere gradients on LAD-GNSS users, the worst differential range error is first tabulated using exhaustive search over combinations of the simulation parameters in the presence of the ground CCD monitor and DSIGMA. Based on the table of the worst differential range error, the performance of the geometry screening monitor is evaluated based on the geometry simulation.

5.1 Worst-case Differential Range Error Due to Anomalous Ionosphere Gradient

This section describes how we model the worst differential errors in range domain caused by anomalous ionosphere gradients. In LAD-GNSS, the code-carrier smoothing is processed to mitigate the receiver noise and multipath on both airborne and ground receivers. However, code-carrier divergence due to ionospheric gradients induces bias into the code-carrier smoothing. Such errors in carrier-smoothed code become larger as either a smoothing time or a magnitude of gradient becomes larger. In LAD-GNSS, both the aircraft and the ground facility use a time constant of 30 seconds for the code-carrier smoothing filter in the presence of anomalous ionosphere gradient.

The resulting differential range errors can be computed by taking the difference between the smoothed ionospheric delay measurements observed from aircraft and the ground facility. Undetected ionosphere-induced differential range errors are then generated for a hypothetical UAV flies in the presence of the modeled ionospheric gradient. The anomalous ionosphere gradient is modeled with several parameters including magnitude, width, direction and relative speed of an ionospheric front with respect to a satellite IPP observed from ground facility. Moving direction of the UAV and the initial position of both the UAV and the front are also included in the simulation parameters. The simulation parameters and those ranges are listed in Table 2. For a given set of simulation parameters, the differential range errors and the test statistic of the monitors are computed every epoch as the ionospheric front and the UAV advances. When the UAV finally reaches at the evaluation point, the differential range error either detected or undetected is recorded. The three different UAV operation scenario is considered, which is summarized in Table 3. The evaluation point is assumed to be 5, 10 and 20 km apart from the ground facility respectively. The horizontal speed of UAV is set at 15 m/s in Coverage I, and 45 m/s in Coverage II, because the speed limit of UAV is constrained to be less than 45 m/s according to FAA regulations.

Table 2. Simulation parameters

Parameter	Min.	Max.	Step Size
Δv (m/s)	-100	350	10
g (mm/km)	200	500	20
W (km)	25	75	25
α (deg)	-90	90	15
β (deg)	-90	90	10

Table 3. Summary of the operating scenario

Scenario	Evaluation Point (km)	Horizontal speed of UAV (m/s)	Coverage
(a)	5	15	I
(b)	10	45	II
(c)	20	45	II

For the monitor threshold, the MDE is taken in order to account for the effect of the noise, so that the MDE values for the ground CCD monitor and DSIGMA is 0.08217 m/s

and 2.02 m respectively. The test statistic of DSIGMA is only generated when the UAV is within the maximum coverage of 20 km.

Figure 7 depicts an example of the results of the worst differential range error in terms of the relative speed of the front. The worst differential range error curves varying with the magnitude of ionosphere gradients from 200 mm/km to 500 mm/km are searched based on the exhaustive simulation over all of the possible combinations of the ionosphere parameters for each given Δv . The constant model (black line) is proposed to conservatively cover the simulated worst range error curves for all the conditions. Since the single-impact scenario is only considered in this analysis, the model of the differential range error does not need to be expressed as a function of Δv . The result of the worst differential range error for the three UAV operation scenario is summarized in Table 4.

Table 4. The maximum of the worst differential range error for each of three UAV operation scenario

Operation scenario	(a)	(b)	(c)
Maximum of range error (m)	3	6.5	11.5

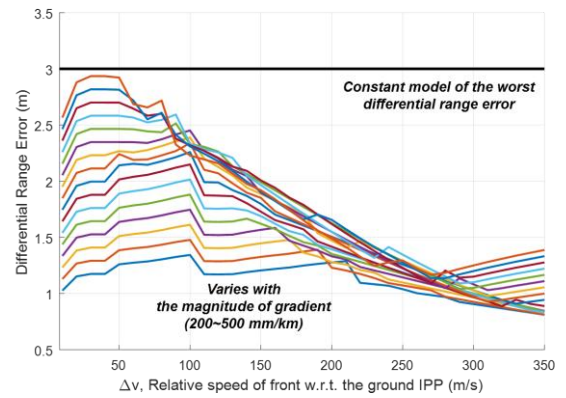


Figure 7. Simulated results of the worst differential range error when the evaluation point is set at 5 km apart from the ground facility and the horizontal speed of UAV is 15 m/s.

5.2 Vertical Position Error due to Ionosphere Gradient

The analytical model of ionosphere-induced differential range error was introduced in Section 5.1. Using this model, we examine the largest potential errors resulting from the worst case scenarios. In order to evaluate the performance of the simplified geometry screening monitor, simulation is conducted to compute the worst-case undetected vertical ionospheric errors by assuming the worst case SV-user geometry and parameter combinations. In this paper, single-satellite impact is assumed to be the worst case scenario instead of double-satellite impact mentioned in [12], because the double-impact scenario is quite conservative assumption for the UAV applications.

For a given single satellite i , a vertical ionospheric error is defined as:

$$IEV_i = |S_{vert,i} \varepsilon_i| \quad (21)$$

where $S_{vert,i}$ and ε_i denote the vertical position component of the weighted-least squares projection matrix S and the ionosphere-induced range error for satellite i respectively. The ionosphere gradient is assumed to move in the worst direction, so the resulting Δv is always to be small. Thus, ε_i is always taken as its maximum in the single-impact scenario.

Among IEVs computed for all of the individual satellites in view, the largest IEV is taken as the Maximum IEV (MIEV). The MIEV is then compared to the limit of the allowable range error due to anomalous ionosphere gradient, TEL. In this paper, we can derive the TEL using Eq. (5) and (6). σ_{TSE} is calculated as the square root sum of fault-free navigation sensor error (NSE) and the flight technical error (FTE). σ_{NSE} is derived from Vertical Alert Limit (VAL) of 10 m, and σ_{FTE} is set at 1.5 m. The obstacle clearance level should be carefully determined because it depends on the environment where the UAV is actually flying above, but H_{OCS} is provisionally assumed to be 0 m in Coverage I and 15 m (as high as four-floor buildings) in Coverage II. The resulting TEL for each coverage is 10.8 m and 41.5 m as shown in Figure 8.

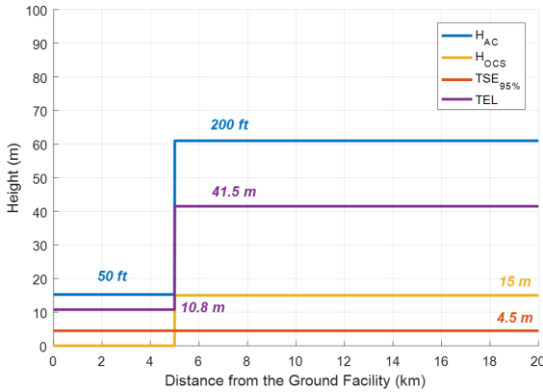


Figure 8. Derivation of TEL for the UAV application

The performance of the geometry screening was evaluated by using the MIEV simulation based on 31 GPS constellation over the course of a day. The satellite geometries are simulated every 5 minutes (which gives 288 epochs for one day). The ground facility is assumed to be located at Ronald Reagan Washington National Airport. The MIEVs are computed for both ideal and pessimistic scenarios. “Ideal” means that the UAV does not lose any satellite in view. However, since the UAV can easily lose satellites due to low elevation noise or intense scintillation, pessimistic scenario of two-satellite outage is also considered. In the pessimistic scenario, the two satellites whose elevation angles are the maximum or the minimum are removed from the all-in-view satellites.

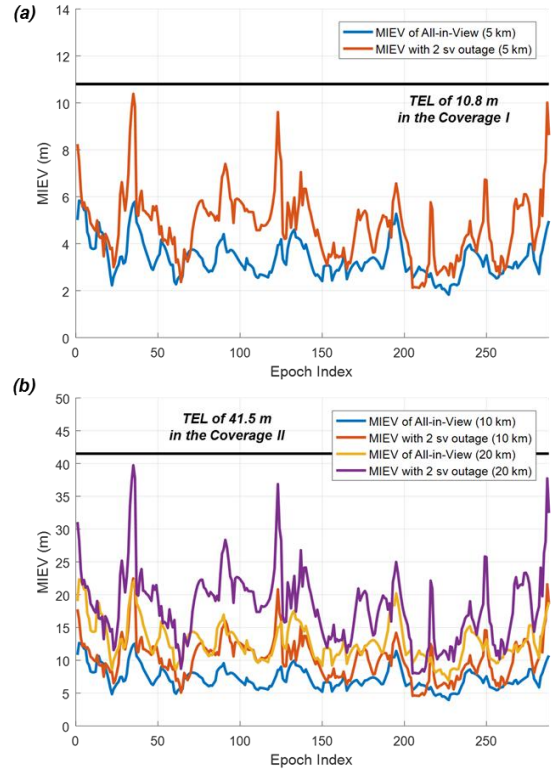


Figure 9. Results of the MIEV simulation for (a) Coverage I and (b) Coverage II.

Figure 9 (a) shows the result of the MIEV simulation for Coverage I when the UAV is operating 5 km apart from the ground facility with a constant speed of 15 m/s. Blue and red curves indicate the MIEV for the ideal and the pessimistic scenarios respectively. The result shows that the MIEV is well below the TEL of 10.8 m in the ideal scenario while the MIEV for pessimistic scenario barely satisfies the operation requirement. Sometimes the MIEV of the ideal scenario is even less than the MIEV of the pessimistic scenario. It happens when the number of tracking satellites is large enough (e.g. 12 for instance). It implies that the number of satellite is not the only factor that contributes to the MIEV.

The MIEV results for Coverage II is shown in Figure 9 (b). The UAV is assumed to fly as fast as the speed limit of 45 m/s, and evaluated at two points which are 10 km and 20 km distant from the ground facility. When the UAV operates within 10 km, the maximum of MIEV for the pessimistic scenario is raised up to about 20 m which sufficiently assures the requirements for Coverage II. However, as the operation distance grows, the MIEV becomes larger. When the UAV is at 20 km, the maximum of the MIEV for the pessimistic condition is slightly below the TEL of 41.5 m. Although the offline simulation results demonstrate that the MIEVs for both Coverage I and II are well below the operation requirements, the MIEV should be monitored in real-time in case of the worse environment like an urban area where it is likely to lose more than two satellites.

6.0 Conclusion

This paper presented the architecture LAD-GNSS for UAVs that communicate ground subsystem in real time. The paper fully analyzed and implemented the LAD-GNSS for UAV, consisting of an operation requirement design, integrity monitoring strategies, and optimal allocation of integrity/continuity budgets. The operation coverages defined for two regions have their respective operating requirements. For the Coverage I, VPL is compared to VAL of 10 m and MIEV is compared to TEL of 10.8 m to determine if the system meets the integrity and continuity requirements for a given operation. The UAV operating Coverage II should meet the VAL of 10 m and TEL of 41.5 m.

In this paper, we derived the PLs for EA and CCD fault case to implement the optimal allocation of integrity/continuity budgets and simulated the VPL under optimal allocation. In the simulation results for a 24-satellite GPS constellation, VPLs of Coverage I and II are reduced by about 8% when implement optimal allocation method. The paper also simulated the worst-case undetected vertical ionospheric errors in order to evaluate the performance of the simplified geometry screening monitor. The results show the MIEVs for Coverage I and II are reasonably well-bounded by the TELs which are pre-determined based on uncertainty of navigation. The NSE and FTE errors highly depend on hardware, environment, and operational conditions and thus need to be evaluated experimentally under various conditions. To support this, future work will include Both the simulation-based analysis and field tests of several types of UAVs in different flight environments and modes.

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